Geforce4 Ti 4200/4600) have a feature that accelerates this anti-aliasing in hardware.

We found almost no differences between the chipsets here. All the tested chips performed well and the Word document showed no visible errors under any of these chipsets. The differences were so minute that we had to stick our noses to the monitor to notice anything worthwhile. All in all, the performance was pretty similar across the board.

Quake III Arena (3D quality)

Introduced by 3dfx in their now defunct Voodoo 5 chipset, Full Screen Anti-Aliasing (FSAA) has now become a major factor in determining 3D quality. Along with the recently introduced feature of anisotropic filtering, FSAA can provide 3D images that are crisp, clear and free of artefacts. Of course, not all the chipsets tested sported these features and thus the quality of their output was not up to par.

The TNT2 M64 and the Trident T64 fared the worst in this test. The image quality within *Quake III Arena* looked washed out when running under these chipsets.

One fact was clearly evident after scouring all the images. The entire Radeon chipset series (the VE, 7500 and the 8500) have brilliant 3D image quality. Of the GeForce series, only the GeForce4 managed to score

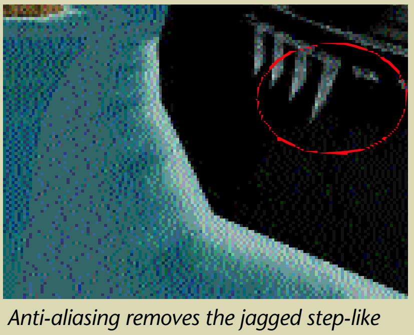
Anisotropic filtering (AF) can make a significant impact on image quality. As we can see in the image on the right, AF has helped in making the floor texture clearer and sharper

Advanced Imaging Features

Be it nVidia with its nFiniteFX engine or ATi with its Truform technology, great efforts are taken by the chipset makers to incorporate features that can enhance image quality without sacrificing on performance.

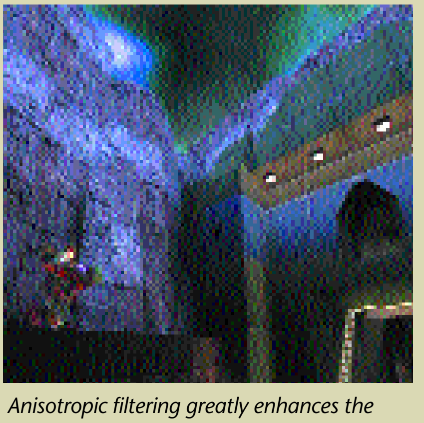
New features like Environment Mapped Bump Mapping (EMBM), Full Screen Anti-Alaising (FSAA), Programmable Vertex and Pixel Shaders, 3D Textures, Shadow Buffers, are promising to deliver the holy grail of computer graphics—true-to-life realism.

Anti-Aliasing (AA): This is an important feature which is incorporated in the new generation chipsets and which greatly enhances image quality. Simply put, AA reduces the jagged edges that are visible on the textures comprising a 3D game. Enabling AA reduces frame rates considerably, but the trade-off is excellent picture quality.



Anti-aliasing removes the jagged step-like artefacts along the edges of game objects

Anisotropic filtering: This serves to reduce texture artefacting and blurring and greatly enhances the image quality with the help of advanced texture filtering techniques. This feature is seen in the latest chipsets such as the GeForce3/4 and the Radeon 8500. Anisotropic filtering in tandem with AA provides excellent, crisp, artefact-free visuals, albeit at a frame rate loss.



Anisotropic filtering greatly enhances the fidelity of an in-game texture. As seen here, the walls sport sharp colours and clean shadows, thanks to this feature

Bump mapping: This is a visual trick that attempts to simulate a texture's roughness or smoothness, depending on how the texture reflects lights. Therefore, you can now have a pond of water with life-like ripple and wave effects. The common techniques used for bump mapping are Emboss Bump Mapping, DOT3 Bump Mapping, Environment Mapped Bump Mapping and True, Reflective Bump Mapping.

Bump mapping has been used in this game to give wave and ripple effects to the water

Programmable shaders: The biggest breakthrough in graphics technology would have to be the introduction of programmable shaders in graphics chipsets, which give developers complete freedom to create custom special effects for their games. Therefore, next generation games will feature eye-candy such as underwater refractions, facial expressions, realistic skin and hair that will make game characters and environments look more true to life than ever before!

Pixel and vertex shaders will enable game developers to create highly-detailed characters like Wolfman here